MATH 128a: Homework #6

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Due by: **Wednesday, October 15, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 2.3

Solve problems 2, 6, 8, 14ac(18ac), 18(16) 30(19) of Section 2.3 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

You can use the Matlab codes on bcourses for problems 6, 8, 14, 18 if useful.

Answer of Problem 1

Exercise 2.3.2. Let $f(x) = -x^3 - \cos x$ and $p_0 = -1$. Use Newton's method to find p_2. Could $p_0 = 0$ be used?

Solution. We have $f'(x) = -3x^2 + \sin x$. So $p_1 = -1 - \frac{f(-1)}{f'(-1)} = -1 - \frac{1-\cos(-1)}{-3+\sin(-1)} \approx -0.880333$ and $p_2 = -0.880333 - \frac{f(-0.880333)}{f'(-0.880333)} \approx -0.865684$.
$p_0 = 0$ could not be used for Newton's method because $f'(0) = 0$.

Exercise 2.3.6. and Exercise 2.3.8. Use Newton's method and the secant method to find solutions accurate to within 10^{-5} for the following problems. (See the data matrix in the solutions for the problems.)

Solution. MATLAB R2025b introduced a built-in for converting symbolic expressions into function handles. This allows us to automate the differentiation of functions. Put the following in a script under the same directory as `newton_1d.m` and `secant.m` to complete all parts simultaneously. Be sure to install the Symbolic Math Toolbox through the Add-On Explorer.

```
syms x;

data = {
    exp(x) + 2^(-x) + 2*cos(x) - 6, [1, 2];
    log(x - 1) + cos(x - 1), [1.3, 2];
    2*x*cos(2*x) - (x - 2)^2, [2, 3];
```

```
    2*x*cos(2*x) - (x - 2)^2, [3, 4];
    (x - 2)^2 - log(x), [1, 2];
    (x - 2)^2 - log(x), [exp(1), 4];
    exp(x) - 3*x^2, [0, 1];
    exp(x) - 3*x^2, [3, 5];
    sin(x) - exp(-x), [0, 1];
    sin(x) - exp(-x), [3, 4];
    sin(x) - exp(-x), [6, 7];
};

cellfun(@process, num2cell(data, 2));

function process(row)
    [f, interval] = deal(row{:});
    df = diff(f);
    mf = matlabFunction(f);
    mdf = matlabFunction(df);
    sf = char(f);

    a = interval(1);
    b = interval(2);
    x0 = (a + b)/2;

    fprintf('f(x) = %s\n', sf);
    fprintf('interval = [%g, %g]\n', a, b);

    epsilon = 1e-5;
    N = 100;
    x = newton_1d(mf, mdf, x0, epsilon, N);
    fprintf('x_N = %.5f\n', x);
    [x, n] = secant(mf, a, b, epsilon, N);
    fprintf('x_S = %.5f\n', x);
    fprintf('\n');
end
```

This should print out:

```
f(x) = 2*cos(x) + exp(x) + 1/2^x - 6
interval = [1, 2]
x_N = 1.82938
x_S = 1.82938

f(x) = cos(x - 1) + log(x - 1)
interval = [1.3, 2]
x_N = 1.39775
x_S = 1.39775
```

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```
f(x) = 2*x*cos(2*x) - (x - 2)^2
interval = [2, 3]
x_N = 2.37069
x_S = 2.37069

f(x) = 2*x*cos(2*x) - (x - 2)^2
interval = [3, 4]
x_N = 3.72211
x_S = 3.72211

f(x) = (x - 2)^2 - log(x)
interval = [1, 2]
x_N = 1.41239
x_S = 1.41239

f(x) = (x - 2)^2 - log(x)
interval = [2.71828, 4]
x_N = 3.05710
x_S = 3.05710

f(x) = exp(x) - 3*x^2
interval = [0, 1]
x_N = 0.91001
x_S = 0.91001

f(x) = exp(x) - 3*x^2
interval = [3, 5]
x_N = 3.73308
x_S = 3.73308

f(x) = sin(x) - exp(-x)
interval = [0, 1]
x_N = 0.58853
x_S = 0.58853

f(x) = sin(x) - exp(-x)
interval = [3, 4]
x_N = 3.09636
x_S = 3.09636

f(x) = sin(x) - exp(-x)
interval = [6, 7]
x_N = 6.28505
x_S = 6.28505
```

Exercise 2.3.14(18). The function $f(x) = tan\pi x - 6$ has a zero at $\pi^{-1} \arctan 6 \approx 0.447431543$.

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Let $p_0 = 0$ and $p_1 = 0.48$. Use ten iterations of bisection method and secant method to approximate this root. Which method is most successful and why?

Solution. You can run this from the command window:

```
>> f = @(x) tan(pi*x) - 6;
>> [x, n] = bisection(f, 0, 0.48, 1e-5)
x = 0.4474
n = 16
>> [x, n] = secant(f, 0, 0.48, 1e-5, 1000)
x = NaN
n = 564
```

It is clear that bisection method outperforms secant method for this problem. This is because the graph of f has a sharp ascension near p_1. This affects the secant method, but it doesn't affect the bisection method, since it updates independently of the actual value of the function and only depends on its sign.

Exercise 2.3.18(16). Use Newton's method to solve

$$\frac{1}{2} + \frac{1}{4}x^2 - x\sin x - \frac{1}{2}\cos 2x = 0$$

within 10^{-5} accuracy. Explain why the result is unusual for Newton's method.

Solution. You can put the following script in the same directory as `newton_1d.m` and run it.

```
syms x;

f = 1/2 + x^2/4 - x*sin(x) - cos(2*x)/2;
df = diff(f);

mf = matlabFunction(f);
mdf = matlabFunction(df);

epsilon = 1e-5;
N = 1000;

for x0 = [pi/2, 5*pi, 10*pi]
    fprintf('x0 = %.5f\n', x0);
    [x, n] = newton_1d(mf, mdf, x0, epsilon, N);
    fprintf('x = %.5f\n', x);
    fprintf('n = %d\n', n);
    fprintf('\n');
end
```

This gives you:

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```
x0 = 1.57080
x = 1.89549
n = 15

x0 = 15.70796
x = 1.89549
n = 19

x0 = 31.41593
x = -1.89550
n = 249
```

This indicates uncharacteristically slow convergence for Newton's method. The reason is immediate from rewriting the function.

$$f(x) = \frac{1}{2} + \frac{1}{4}x^2 - x\sin x - \frac{1}{2}\cos 2x = \frac{1}{4}(x - 2\sin x)^2$$

Because the square, the function always vanishes to at least second order, meaning that $f'(x)$ is not bounded away from 0 when $f(x)$ vanishes. The graph of the function flattens out at the root, so Newton's method converges more slowly than expected.

Exercise 2.3.30(19). The iteration equation for the secant method can be written in the simpler form

$$p_n = \frac{f(p_{n-1})p_{n-2} - f(p_{n-2})p_{n-1}}{f(p_{n-1}) - f(p_{n-2})}.$$

Explain why, in general, this iteration equation is likely to be less accurate than the one given in Algorithm 2.4.

Solution. In the usual secant method, $p_n = p_{n-1} - \frac{f(p_{n-1})(p_{n-1}-p_{n-2})}{f(p_{n-1})-f(p_{n-2})}$, roundoff errors in computing $f(p_{n-1})$, $(p_{n-1} - p_{n-2})$ and $f(p_{n-1}) - f(p_{n-2})$ cause loss of relative accuracy in the *correction* to p_{n-1}. But this correction is likely to be much smaller than p_{n-1}, so having only, say, two correct digits of the corrector can cause, say, the 11th and 12th digit to be fixed in updating p_{n-1} to obtain p_n. In the formula above, the new p_n will not have more correct digits than the denominator, which may suffer severely from cancellation error (especially as the terms converge to the correct answer and $f(p_{n-1})$ and $f(p_{n-2})$ are both close to zero.)

Problem 2: Problems from the textbook, chapter 2.4

Solve problems 1b, 6b, 10 of Section 2.4 of the textbook.

6b: Show that p_n converges at order $\alpha = 1$, but not linearly, since $\lambda = 1$ in Definition 2.7.

(If the problem has a different number in the 9th edition book, it is in parentheses)

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Answer of Problem 2

Exercise 2.4.1. Use Newton's method to find solutions accurate to within 10^{-5} for the following problems.

$$\text{b.} \quad \cos(x+\sqrt{2}) + x(x/2+\sqrt{2}) = 0, \qquad (-2 \le x \le -1).$$

Solution. Let $f(x) = \cos(x+\sqrt{2}) + x(x/2+\sqrt{2})$. Then

$$f'(x) = -\sin(x+\sqrt{2}) + (x/2+\sqrt{2}) + x \cdot \frac{1}{2} = x + \sqrt{2} - \sin(x+\sqrt{2}).$$

So to implement Newton's method, we use fixed-point iteration with the function

$$g(x) = x - \frac{f(x)}{f'(x)} = x - \frac{\cos(x+\sqrt{2}) + x(x/2+\sqrt{2})}{x+\sqrt{2}-\sin(x+\sqrt{2})}. \qquad (*)$$

Since the interval is $[-2,-1]$, we will start with initial value -1.5. Then the sequence of approximations generated by Newton's method is as follows:

$$\begin{aligned}
&-1.5, -1.47855, -1.46247, -1.4504, -1.44135, -1.43457, -1.42948,\\
&-1.42566, -1.4228, -1.42065, -1.41904, -1.41784, -1.41693, -1.41625,\\
&-1.41574, -1.41536, -1.41507, -1.41486, -1.4147, -1.41459, -1.41451,\\
&-1.41449, -1.41445, -1.41445.
\end{aligned}$$

The slow convergence rate is due to $f(x)$ having a zero of multiplicity 4 at $x = -\sqrt{2}$. To see this, note that $f(s-\sqrt{2}) = \cos(s) - 1 + s^2/2 = O(s^4)$, so $f(x) = O(|x+\sqrt{2}|^4)$. In fact, Newton's method fails to converge to the requested accuracy due to roundoff error in double-precision. (The answer is -1.41421.)

Exercise 2.4.6. Show that the following sequences converge linearly to $p = 0$. How large must n be before we have $|p_n - p| \le 5 \times 10^{-2}$?

$$\text{b.} \quad p_n = \frac{1}{n^2}, \qquad (n > 1).$$

Solution. It is easy to show that $\left\langle \frac{1}{n^2} \right\rangle \to 0$ by a direct argument from the definition. Let $\epsilon > 0$ choose $N = \lceil \frac{1}{\sqrt{\epsilon}} \rceil$. Let $n \in \mathbb{N}$ with $n > N$. Then $|p_n - 0| = |\frac{1}{n^2}| < \frac{1}{N^2} \le \epsilon$.

(The convergence is actually *not* linear. We would have to show that for some constant $\lambda < 1$ we have $\lim_{n\to\infty} \frac{|p_{n+1}-0|}{|p_n-0|} = \lambda$. In our case, we have $\frac{|1/(n+1)^2-0|}{|1/n^2-0|} = \frac{n^2}{(n+1)^2} = \left(1 - \frac{1}{n+1}\right)^2$, which approaches 1 as $n \to \infty$. So the convergence is sub-linear since $\lambda = 1$.)
To get the error less than $\frac{5}{100} = \frac{1}{20}$, we need to have $|\frac{1}{n^2} - 0| < \frac{1}{20}$, so we need $n^2 > 20$, which requires $n \ge 5$.

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Exercise 2.4.10. Suppose p is a zero of multiplicity m of f, where $f^{(m)}$ is continuous on an open interval containing p. Show that the following fixed-point method has $g'(p) = 0$:

$$g(x) = x - \frac{mf(x)}{f'(x)}$$

Solution. If $m = 1$, we assume additionally that $f \in C^2[a, b]$ and note that

$$g'(x) = f(x)f''(x)/f'(x)^2 \to f(p)f''(p)/f'(p)^2 = 0,$$

where we used $f(p) = 0$ and $f'(p) \neq 0$ when $m = 1$.
If $m \geq 2$, then since p has multiplicity m, we can write $f(x) = (x-p)^m q(x)$ where $\lim_{x\to p} q(x) \neq 0$. Using this form to take the derivative of f we get

$$\begin{aligned} f'(x) &= m(x-p)^{m-1}q(x) + (x-p)^m q'(x), \\ f''(x) &= m(m-1)(x-p)^{m-2}q(x) + 2m(x-p)^{m-1}q'(x) + (x-p)^m q''(x). \end{aligned}$$

We also note that

$$\frac{d}{dx}\left(\frac{f(x)}{f'(x)}\right) = 1 - \frac{f(x)f''(x)}{f'(x)^2} = 1 - \frac{[f(x)/(x-p)^m][f''(x)/(x-p)^{m-2}]}{[f'(x)/(x-p)^{m-1}]^2}.$$

Taking the limit as $x \to p$, this becomes $1 - \frac{[q(p)][m(m-1)q(p)]}{[mq(p)]^2} = 1 - \frac{m-1}{m} = \frac{1}{m}$. Here we used the fact that $(x-p)q'(x)$ and $(x-p)^2q''(x)$ approach 0 as $x \to p$, as shown in `Lec11.pdf`. Alternatively, we can use Taylor's theorem to expand $f(x)$, $f'(x)$ and $f''(x)$, making use of $f^{(j)}(p) = 0$ for $0 \leq j \leq m-1$:

$$\begin{aligned} f(x) &= \frac{1}{m!}f^{(m)}(\xi_0(x))(x-p)^m, \\ f'(x) &= \frac{1}{(m-1)!}f^{(m)}(\xi_1(x))(x-p)^{m-1}, \\ f''(x) &= \frac{1}{(m-2)!}f^{(m)}(\xi_2(x))(x-p)^{m-2}. \end{aligned}$$

This gives

$$\frac{f(x)f''(x)}{f'(x)^2} = \frac{(m-1)!^2}{m!(m-2)!} \cdot \frac{f^{(m)}(\xi_0(x))f^{(m)}(\xi_2(x))}{f^{(m)}(\xi_1(x))^2} \to \frac{m-1}{m} \text{ as } x \to p,$$

just as before. Finally, we have

$$g'(x) = 1 - m\frac{d}{dx}\left(\frac{f(x)}{f'(x)}\right) \overset{x\to p}{\longrightarrow} 1 - m\frac{1}{m} = 0.$$

Remark: this method breaks down unless $f(x)$ and $f'(x)$ can be computed with at least a few correct digits near the zero p. For example, in Exercise 2.4.1 above, if we replace (∗) with

$$g(x) = x - 4\frac{f(x)}{f'(x)} = x - 4\frac{\cos(x+\sqrt{2}) + x(x/2+\sqrt{2})}{x+\sqrt{2}-\sin(x+\sqrt{2})},$$

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the sequence of iterates is

$$-1.5,\ -1.41420304,\ 0.8722888,\ -1.62895419,\ -1.414048413,$$
$$-1.4134569,\ -1.4142071344,\ 18.650009, -23.30858,\ \ldots$$

Note that $p_1 = g(1.5)$ is much more accurate than it was for Newton's method (∗). However, each time the sequence gets close to $p = -\sqrt{2} = -1.4142135624$, roundoff error eliminates all correct digits of the correction, and the next iterate is nearly random. But if instead we use the formulas

$$f(x) = \begin{cases} [\cos(s) - 1 + s^2/2]_{s=x+\sqrt{2}}, & |x+\sqrt{2}| > .01, \\ [s^4/24 - s^6/720 + s^8/40320]_{s=x+\sqrt{2}}, & |x+\sqrt{2}| \le .01, \end{cases}$$
$$f'(x) = \begin{cases} [s - \sin(s)]_{s=x+\sqrt{2}}, & |x+\sqrt{2}| > .01, \\ [s^3/6 - s^5/120 + s^7/5040]_{s=x+\sqrt{2}}, & |x+\sqrt{2}| \le .01, \end{cases}$$

where we use Taylor series to avoid unnecessary cancellation of digits as $x \to \sqrt{2}$, then this variant of Newton's method converges very quickly:

$$-1.5,\ -1.41420304,\ -1.4142135623730951,\ -1.4142135623730951.$$

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